

Multiple Linear Regression :

Ex: Response (y): IQ

Predictor : Brain size, Height, weight

- relationship between response and each of the predictor and how the predictors are related.

Scatter plot matrix

- linear in parameters. The model, $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$, is a multiple linear regression model.

Possible questions :

- 1) Which, if any, predictors - brain size, height or weight - explain some of the variations in intelligence scores. (each slope parameter could be zero)
- 2) What is the effect of brain size on IQ, after taking into account height and weight? (calculate and interpret a confidence interval for the brain size slope parameter)
- 3) Predict IQ given brain size, height and weight.

Model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_{p-1} x_{i,p-1} + \varepsilon_i, \quad i = 1(1)n$$

Observation Number	Response y	Predictors				
		x_1	x_2	$\dots$	$\dots$	x_{p-1}
1	y_1	x_{11}	x_{12}	$\dots$	$\dots$	x_{1p-1}
2	y_2	x_{21}	x_{22}	$\dots$	$\dots$	x_{2p-1}
$\vdots$	$\vdots$	$\vdots$				
n	y_n	x_{n1}	x_{n2}	$\dots$	$\dots$	x_{np-1}

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p-1} \\ 1 & x_{21} & x_{22} & \dots & x_{2p-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np-1} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_{p-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

Assumptions:

- **Linearity:** The mean of response variable is a linear combination of the parameters and predictor variables.
- **Constant Variance (Homoscedasticity):** The errors, ε_i , at each set of values of the predictors ($x_{1i}, x_{2i}, \dots$) have equal variance.
In simple terms, it means the spread of the errors should be relatively uniform, regardless of the value of the predictor.
- **Independent:** The errors are independent.
- **Normally Distributed:** The errors are Normally distributed.
- **$\text{Rank}(X) = k$:** The independent variables are not highly correlated to each other.

Interpretation of the Model Parameters :

- Each β parameter represents the change in the mean response, $E(y)$, per unit increase in the associated predictor variable when all the other predictors are held constant.
For example, β_1 represents the estimated change in the mean response, $E(y)$, per unit increase in x_1 when $x_2, x_3, \dots, x_{p-1}$ are held constant.
- The intercept term, β_0 , represents the estimated mean response, $E(y)$, when all predictors $x_1, x_2, \dots, x_{p-1}$ are all zero. (which may or may not have practical meaning)

Model Adequacy Checking :

- checking the linear relationships between study and explanatory variables :

Case 1: One explanatory variable :

Scatter plot. If the scatter diagram shows a linear trend, it indicates that the relationship between y and x is linear. If the pattern is not linear, then it suggests that the relationship between y and x is nonlinear.

Case 2: More than one explanatory variable :

- scatter plot matrix.

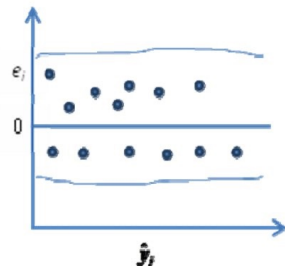
- create a scatter plot with residuals, e_i , on the vertical axis and the fitted values, $\hat{y}_i$, on the horizontal axis and visual assess whether:

- the (vertical) average of the residuals close to 0 as we scan the plot from left to right.
- the (vertical) spread of the residuals remains approximately constant as we scan the plot from left to right
- there is no excessively outlying points.

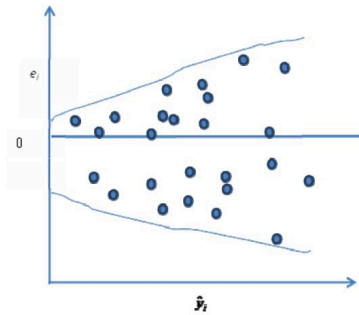
- If the data observations were collected over time (or space) create a scatterplot with the residuals e_i , on the vertical axis and the time (or space) on the horizontal axis and visual assess whether there is no systematic non random pattern.
 - violation may suggest the need of time series model.

Plot of residuals against the fitted values :

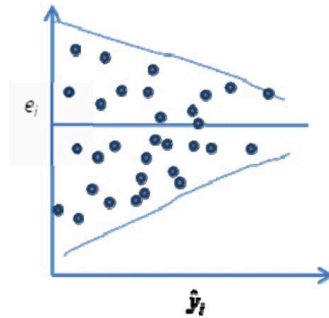
- a) If the plot is such that the residuals can be contained in **horizontal band** fashion (and residuals fluctuates more or less in a random manner inside the band) then there is no visible model defects.



- b) If the plot is such that the residuals can be contained in an outward opening funnel then such pattern indicates the variance of errors is not constant, but it is an increasing function of y .



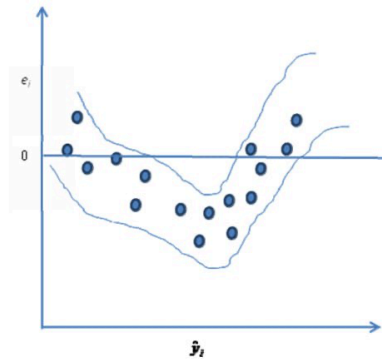
- c) If plots are such that the residuals can be accommodated in an inward opening funnel, then such a pattern indicates the variance of errors is not constant, but it is a decreasing function of y .



- d) If the plot is such that the residuals can be accommodated inside a double bow, then such a pattern indicates that the variance of errors is not constant but y is proportion between 0 and 1. The assumed relationship between y and X 's is nonlinear.

[The usual approach to deal with such inequality of variances is to apply a suitable transformation to either the explanatory variables or study variable or use the method of weighted least square. In practice, transformations on study are generally employed to stabilize the variance]

e) If the plot is such that the residuals are contained inside a **curved plot**, then it indicates nonlinearity. The assumed relationship between y and X 's is non-linear. This could also mean some other predictors are needed in the model. Transformation on predictors may be helpful in these cases.





Multicollinearity :

A basic assumption in MLR is that the rank of the matrix of observations on explanatory variables is the same as the number of explanatory variables. In other words such a matrix is of full column rank. This in turn, implies that all the explanatory variables are independent, i.e, there is no linear relationship among the explanatory variables.

In many situations in practice the explanatory variables may not remain independent due to various reasons. The situations where the explanatory variables are highly correlated is referred to as multicollinearity.

Sources of multicollinearity :

1. Method of data collection: It is expected that the data is collected over whole cross-section of the variables. It may happen that data is collected over a subspace of explanatory variable where the variables are linearly dependent.

2. Existence of identities or definitional relationships: There may exist some relationships among the variables which may be due to the definition of variables or identity relation among them. For example: if data is collected on the variables like income, saving and expenditure, then

$$\text{income} = \text{Saving} + \text{expenditure}$$

3. Imprecise Formulation of model :

The formulation of the model may unnecessarily be complicated. For example, the quadratic (or polynomial) terms or cross-product terms may appear as explanatory variables.

For example, let there be 3 variables X_1, X_2 and X_3 . Suppose their cross-product terms X_1X_2, X_2X_3 and X_1X_3 are also added.

4. An over-determined model:

Sometimes a large no. of variables are included in the model to make it more realistic. Consequently the number of observations become smaller than the number of explanatory variables.

Ex: Such a situation may arise in medical research where the number of patients may be small but the information is collected on a large no. of variables.

Consequences of multicollinearity:

1. The OLSE remains an unbiased estimator of β , but its sampling variance becomes very large. So the OLSE becomes imprecise and the property of BLUE does not hold anymore.
- ② when the predictor variables are correlated, the estimated regression coefficient of any one variable depends on which other predictor variables are included in the model.

Ex: $Y = BP$, $X_2 = \text{weight}$, $X_3 = \text{BSA (Body surface area)}$

Here's the relevant portion of the table:

Variables in model	weight b_2	BSA b_3
x_2	1.20	---
x_3	---	34.4
x_2, x_3	1.04	5.83

Note that, depending on which predictors we include in the model, we obtain wildly different estimates of the slope parameter for $x_3 = \text{BSA}$!

- If $x_3 = \text{BSA}$ is the only predictor in the model, we claim that for every additional one square meter increase in body surface area (BSA), BP increase by 34.4 mm Hg.
- On the other hand if $x_2 = \text{weight}$ and $x_3 = \text{BSA}$ are both included in the model, we claim that for every additional one square meter increase in body surface area, holding weight constant, BP increase only 5.83 Hg.

This is a huge difference!

The high correlation between the two predictors is what causes the large discrepancy.

When interpreting $b_3 = 34.4$ in the model that excludes $x_2 = \text{weight}$, keep in mind that when we increase $x_3 = \text{BSA}$ then $x_2 = \text{weight}$ also increases and both factors are associated with increase blood pressure.

However, when interpreting $b_3 = 5.83$ in the model that includes $x_2 = \text{weight}$, we keep $x_2 = \text{weight}$ fixed, so the resulting increase in blood pressure is much smaller.

The fantastic thing is that even predictors that are not included in the model, but are highly correlated with the predictors in our model, can have an impact! For example, consider a pharmaceutical company's regression of *territory sales* on *territory population* and *per capita income*. One would, of course, expect that as the population of the territory increases, so would the sales in the territory. But, contrary to this expectation, the pharmaceutical company's regression analysis deemed the estimated coefficient of territory population to be negative. That is, as the population of the territory increases, the territory sales were predicted to decrease. After further investigation, the pharmaceutical company determined that the larger the territory, the larger to the competitor's market penetration. That is, the competitor kept the sales down in territories with large populations.

- ② when predictor variables are correlated, the precision of the estimated regression coefficients decreases as more predictor variables are added to the model. The standard error for the estimated parameters increases as we increase covariates in the model. Standard errors are used in the calculation of the confidence intervals for the slope parameters. That is, increased standard errors of the estimated slopes lead to wider confidence intervals and hence less precise estimates of the slope parameters.
- ③ when predictors are correlated, hypothesis tests for $\beta_k = 0$ may yield different conclusions depending on which predictor variables are in the model.

Note :- High multicollinearity among predictor variables does not prevent good, precise predictions of the response within the scope of the model.

If the primary purpose of your regression analysis is to estimate a mean response or predict a new response y , you don't have to worry much about multicollinearity.

Detecting Multicollinearity Using Variance Inflation Factors :

Some common methods used for detecting multicollinearity include :

- Estimates of coefficients varying from model to model.
- The t -tests for each of the individual slopes are non-significant ($P > 0.05$), but the overall F -test for testing all of the slopes is simultaneously 0 is significant.
- The correlation among pairs of variables are large. (Limiting)

It is possible that pairwise correlations are small and yet a linear dependence exists among three or even more variables, for example $x_3 = 2x_1 + 5x_2 + \text{error}$.

VIF:- variances of the estimated coefficients are inflated when multicollinearity exists. So, the VIF for the estimated coefficient b_k - denoted VIF_k - is just the factor by which variance is inflated.

For the model in which x_k is the only predictor

$$y_i = \beta_0 + \beta_1 x_{ik} + \varepsilon_i$$

It can be shown that variance of estimated coefficient b_k is :

$$\text{Var}(b_k)_{\min} = \frac{\sigma^2}{\sum_{i=1}^n (x_{ik} - \bar{x}_k)^2}$$

Note that we add "min" in order to denote that it is the smallest variance can be.

Let's consider model with correlated predictors

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \dots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$$

Now again, if some of the predictors are correlated with predictor x_k , then the variance of b_k is inflated.

It can be shown that variance of b_k is:

$$\text{Var}(b_k) = \frac{\sigma^2}{\sum_{i=1}^n (x_{ik} - \bar{x}_k)^2} \times \frac{1}{1 - R_k^2}$$

where R_k^2 is the R^2 value obtained by regressing k th predictor on the remaining predictors.

Of course, the greater linear dependence among the predictors x_k and other predictors, the larger the R_k^2 value.

And as the above formula suggests, the larger the R_k^2 value, the larger the variance of b_k .

$$\frac{\text{Var}(b_k)}{\text{Var}(b_k)_{\min}} = \frac{1}{1 - R_k^2}$$

The above quantity is deemed the variance inflation factor (VIF) for the k th predictor. That is,

$$\text{VIF}_k = \frac{1}{1 - R_k^2}.$$

How do we interpret the VIF for a regression model?

→ It measures how much the variance of the estimated regression coefficient b_k is inflated by existence of correlation among the predictor variables in the model.

A VIF of 1 means that there is no correlation between the k th predictor and the remaining predictor variables and hence variance of b_k is not inflated at all.

$\text{VIF} > 4 \rightarrow$ warrant investigation

$> 10 \rightarrow$ serious multicollinearity requiring correction.



Normality

Q-Q (Quantile - Quantile Plot)

A graphical tool used to compare two probability distributions by plotting their quantiles against each other.

It's most common use is to compare the distribution of the sample dataset against a theoretical distribution.

Ex: x axis: Theoretical quantiles of $N(0,1)$
y axis: Sample quantiles

- A Q-Q plot helps you instantly check -
 1. Is the Normality assumption valid
 2. If not, how does the data deviate?
(Is it skewed? Does it have heavy tails?)

Q4: How is a Normal QQ Plot constructed?

- A:
 1. **Sort your data:** Take your sample data $x_1, x_2, \dots, x_n$ and sort it in ascending order to get the order statistics $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$.
 2. **Calculate theoretical quantiles:** For each data point $x_{(i)}$, we find the corresponding z-score (theoretical quantile) from the standard normal distribution, $z_{(i)}$, such that the proportion of data below $z_{(i)}$ is equal to a specific probability p_i .
 3. **Choose the probability p_i :** A common method is $p_i = (i - 0.5) / n$, where i is the rank and n is the sample size. This adjusts for the fact that we are estimating population quantiles from a sample. (Other formulas like $i / (n+1)$ are also used).
 4. **Plot the pairs:** The QQ plot is a scatter plot where the x-axis is the theoretical quantile $z_{(i)}$ and the y-axis is the sample quantile $x_{(i)}$.

Interpretation :

1. Heavy Tails (Leptokurtic):

- **What it looks like:** A distinct "S" shape. The points are *below* the line on the far left (low values) and *above* the line on the far right (high values).
- **What it means:** Your data has **more outliers** (or more extreme values) than a normal distribution. The tails are "fatter" or "heavier." This is common in financial data.

2. Light Tails (Platykurtic):

- **What it looks like:** A reverse "S" shape. The points are *above* the line on the far left and *below* the line on the far right.
- **What it means:** Your data has **fewer outliers** (or fewer extreme values) than a normal distribution. The tails are "thinner." This might happen if the data is bounded or truncated.

3. Right Skew (Positive Skew):

- **What it looks like:** A  shape (concave up). The points start above the line, dip down, and then rise sharply, ending far above the line.
- **What it means:** Your data is "piled up" on the left and has a long tail of high values to the right. Both the low and high values in your data are "higher" than what a normal distribution would predict.
- **How to fix:** Consider a log transformation or square root transformation.

4. Left Skew (Negative Skew):

- **What it looks like:** A  shape (concave down). The points start below the line, rise up, and then flatten out, ending below the line.
- **What it means:** Your data is "piled up" on the right and has a long tail of low values to the left.
- **How to fix:** Consider a square transformation (x^2) or exponential transformation.

why use Q-Q plot when formal statistical tests for normality exist?

1. Sensitivity to Sample Size:

- **With large datasets ($n > 1000$):** Formal tests become *overly sensitive*. They will detect even tiny, trivial deviations from perfect normality and give you a small p-value (reject H_0), telling you your data is "not normal." In reality, no real-world data is perfectly normal, and the Q-Q plot might show that the deviation is practically insignificant.
- **With small datasets ($n < 30$):** Formal tests have *low power*. They will often fail to detect that the data is non-normal (fail to reject H_0) even when the Q-Q plot shows a clear deviation.

2. Actionable Insight:

- A formal test gives a **binary answer**: "Normal" or "Not Normal."
- A Q-Q plot tells you **how** the data is not normal (e.g., "it's right-skewed with heavy tails"). This is far more useful, as it directly informs your next step (e.g., "I should try a log transform" or "I should use a robust model that isn't sensitive to outliers").

Summary: Formal tests tell you **if** it's normal; Q-Q plots tell you **how** it's not normal.

Q7: How is a Q-Q plot different from a P-P plot?

This is a less common but good "distinction" question.

- **Q-Q Plot (Quantile-Quantile):**
 - Plots: **Sample Quantiles** (values) vs. **Theoretical Quantiles** (values).
 - Question it answers: "Does the 95th percentile value in my data match the 95th percentile value of a normal distribution?"
 - Sensitivity: Better at showing discrepancies in the **tails** (extremes) of the distribution.
- **P-P Plot (Probability-Probability):**
 - Plots: **Sample Cumulative Probability** vs. **Theoretical Cumulative Probability**.
 - Question it answers: "If I pick a value (e.g., 5.0), does the **percentage** of my data below 5.0 match the **percentage** of a normal distribution below 5.0?"
 - Sensitivity: Better at showing discrepancies in the **center** (around the median) of the distribution.

Shapiro-Wilk Test

H_0 : The errors follow a normal distribution

H_1 : The errors don't follow a normal distribution.

$$W = \frac{\left(\sum_{i=1}^n a_i e_{(i)} \right)^2}{\sum_{i=1}^n (e_i - \bar{e})^2}$$

$e_{(i)}$: order statistics of e_i .

a_i : $E[e_{(i)}]$ if $e_1, e_2, \dots, e_n$ random sample from $N(0,1)$.

Intuition :

The most intuitive way to understand the W statistic is to think of it as a **ratio of two different ways to estimate the population variance (σ^2)**.

$W = \frac{\text{A "normality-based" estimate of variance}}{\text{The "standard" estimate of variance}}$

- If your data is **truly normal**: Both estimates will be very similar, so their ratio (W) will be close to 1.
- If your data is **not normal**: The "normality-based" estimate will be a poor-quality estimate and will be *smaller* than the "standard" estimate. This will make the ratio (W) **less than 1**.

• W lies between 0 and 1.

• $W \approx 1$: suggests that data is normal

$W < 1$: The data is likely not normal

Q8: What are the key assumptions and limitations of the Shapiro-Wilk test?

- A:
 - **Sample Size**: It is most reliable for small to medium sample sizes (roughly $3 \leq n \leq 5000$). The test is very powerful for small samples.
 - **Independence**: The data points are assumed to be independent and identically distributed (i.i.d.).
 - **Sensitivity to Sample Size**: For very large samples ($n > 5000$), the test can become overly sensitive, detecting even trivial, practically insignificant deviations from normality and rejecting the null hypothesis. In such cases, the QQ plot is often more informative.

- The Shapiro-Wilk W statistic is essentially a numerical measure of how well your data points fit the straight line on a Q-Q plot.

Q4: How do the components of W relate to the Q-Q plot visualization?

A: Let's break it down:

- **Numerator** $(\sum a_i x_i)^2$: This is the "good variance" - the variance explained by the normal distribution. On the Q-Q plot, this corresponds to how well your points align with the ideal normal line.
- **Denominator** $\sum (x_i - \bar{x})^2$: This is the **total variance** in your data. On the Q-Q plot, this is the overall vertical spread of your points.
- **Ratio W** : The proportion of total variance that can be explained by the normal distribution model.

- Kolmogorov Smirnov Test :-

Test for Constant Error Variance

Breusch-Pagan Test

The **Breusch-Pagan test** (also known as the **Cook-Weisberg score test**) is an alternative to the modified Levene test. Whereas the modified Levene test is a nonparametric test, the Breusch-Pagan test assumes that the error terms are normally distributed, with $E(\epsilon_i) = 0$ and $\text{Var}(\epsilon_i) = \sigma_i^2$ (i.e., nonconstant variance). The σ_i^2 values depend on the horizontal axis quantity (X_i) values in the following way:

$$\log \sigma_i^2 = \gamma_0 + \gamma_1 X_i$$

We are interested in testing the null hypothesis of constant variance versus the alternative hypothesis of nonconstant variance. Specifically, the hypothesis test is formulated as:

$$\begin{aligned} H_0: \gamma_1 &= 0 \\ H_A: \gamma_1 &\neq 0 \end{aligned}$$

This test is carried out by **first regressing the squared residuals on the predictor (i.e., regressing e_i^2 on X_i)**. The **sum of squares resulting from this analysis is denoted by SSR^*** , which provides a measure of the dependency of the error term on the predictor. The test statistic is given by:

$$X^{2*} = \frac{SSR^* / 2}{(SSE/n)^2}$$

where **SSE is from the regression analysis of the response on the predictor**. The p -value for this test is found using a χ^2 distribution with 1 degree of freedom (written as χ_1^2).

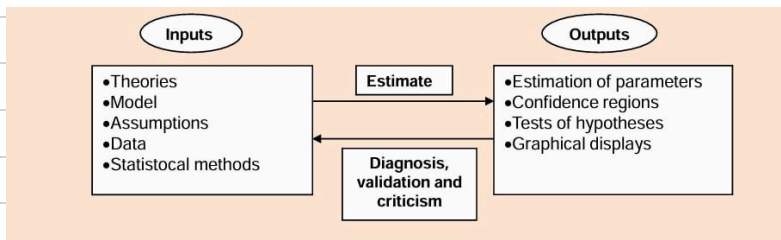
Linear Regression:- A statistical method used to model relationship between a dependent variable and one or more independent variable by fitting a straight line or hyperplane to the data.

Linear Model: Linear in parameters.

Simple Linear Regression: Only one dependent variable.

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i=1, \dots, n$$

Multiple Linear Regression: Two or more dependent variable.

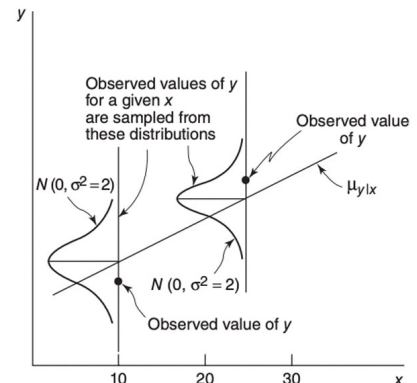


$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i=1, \dots, n$$

$$\varepsilon_i \stackrel{iid}{\sim} (0, \sigma^2)$$

$$E[y_i | x_i] = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}$$

$$V[y_i | x_i] = \sigma^2$$



Vector form: $y_i = x_i^T \beta + \varepsilon_i$

Matrix form: $y = X\beta + \varepsilon$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

$n \times (p+1)$

$$E[y] = X\beta$$

parameter estimation:-

Least Square :-

$$S(\beta) = (y - X\beta)^T (y - X\beta)$$

$$\hat{\beta}_{OLS} = \arg \min_{\beta \in \mathbb{R}^{p+1}} S(\beta) = \arg \min_{\beta \in \mathbb{R}^{p+1}} (y - X\beta)^T (y - X\beta)$$

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

- $(X^T X)^{-1}$ exists if $n > p+1$
no. of observations > no. of predictors

- The columns of X are linearly independent.

$$\hat{y} = X\hat{\beta} = X \underbrace{(X^T X)^{-1} X^T y}_H$$

properties of OLS estimator:-

- $E[\hat{\beta}_{OLS}] = \beta$
- Gauss Markov Theorem:- For the linear regression model

$\hat{\beta}_{OLS}$ is the BLUE.

Estimation of σ^2 :

$$\hat{\sigma}_{ols}^2 = \frac{e^T e}{n-p-1} = \frac{SS_{Res}}{n-p-1}$$

Maximum Likelihood :-

$$\varepsilon \sim N(0, \sigma^2 I_n)$$

$$y \sim N(X\beta, \sigma^2 I_n)$$

$$\hat{\beta}_{MLE} = \hat{\beta}_{ols} = (X^T X)^{-1} X^T y$$

$$\sigma_{MLE}^2 = \frac{(y - X\hat{\beta}_{ols})^T (y - X\hat{\beta}_{ols})}{n} = \frac{e^T e}{n}$$

- $\hat{\beta}_{ols} \sim N_{p+1}(\beta, \sigma^2 (X^T X)^{-1})$

Properties of H matrix :-

$$H = X(X^T X)^{-1} X^T$$

→ H is idempotent

→ H is symmetric

→ H is a projection matrix to the column space of X.

→ $\text{rank}(H) = p+1$

- If the model is such that it has an intercept term, then the matrix $X = [1_n \tilde{X}]$
then $1_n^T e = \sum e_i = 0$.

- The residuals are orthogonal to design matrix.
 $e^T X = 0$.

- $SS_T = (y - \bar{y} 1_n)^T (y - \bar{y} 1_n) = \sum_{i=1}^n (y_i - \bar{y})^2$
 $SS_{Res} = e^T e = (y - \hat{y})^T (y - \hat{y}) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
 $SS_R = (\hat{y} - \bar{y} 1_n)^T (\hat{y} - \bar{y} 1_n) = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

If $X = [1 \ X]$

$\therefore \underbrace{SS_T}_{\text{Total variability in Response}} = SS_{Res} + SS_R$

If $SS_{Res} \gg SS_R$: most of the variability left + unexplained

$SS_{Res} \ll SS_R$: much of the variability in the y 's explained by regressors.

Test for significance of regression

$$H_0: \beta_1 = \beta_2 = \dots = \beta_j = 0$$

$$H_1: \beta_j \neq 0 \text{ for at least one } j$$

$$F_0 = \frac{SSR / p}{SS_{Res} / (n - p - 1)} \sim F_{p, n-p-1}$$

$$F_0 > F_{\alpha; p, n-p-1}$$

R^2 and Adj. R^2

$$R^2 = \frac{SSR}{SS_T}$$

R^2 is the amount of variability in the response that is explained by the regressors.

$R^2 \approx 1$ indicates an excellent fit.

$R^2 \approx 0$ " a not good fit.

$$R^2_{Adj.} = 1 - \frac{SS_{Res} / (n - p - 1)}{SS_T / (n - 1)}$$

- **The Problem with R^2 :** Its value **always increases** every time you add a new variable to the model. It never goes down. This is dangerous because you could add a completely useless, random variable (like 'shoe size' to predict 'stock price'), and your R^2 would still go up, making your model *look* better when it's actually just more complex and overfit.
- **Adjusted R^2** solves this problem. It's a modified version of R^2 that **penalizes the model for adding variables** that don't actually improve its predictive power.

Test on Individual Regression Coefficients

$$H_0: \beta_j = 0 \quad \text{vs} \quad H_1: \beta_j \neq 0$$

$$t_0 := \frac{\hat{\beta}_j}{\sqrt{\hat{\sigma}^2 c_{jj}}} \sim t_{n-p-1}$$

Residual Analysis :-

$$e = y - \hat{y}$$

$$e_i = y_i - \hat{y}_i$$

standardized residual :-

$$d_i = \frac{e_i}{\hat{\sigma}} \quad , \quad i=1, \dots, n$$

$$|d_i| > 3 \Rightarrow \text{outlier}$$

Studentized residual :-

$$r_i = \frac{e_i}{\sqrt{\hat{\sigma}^2 (1 - h_{ii})}} \sim t_{n-p-1}$$

$$|r_i| > 3 \Rightarrow \text{outlier}$$

Press residual :-

$e_{(i)} = y_i - \hat{y}_{(i)} \rightarrow$ How different would the fitted value be without this i th observation.

- An outlier is a data point whose response y does not follow the general trend of the rest of the data.
- A data point has **high leverage** if it has "extreme" predictor x values. With a single predictor, an extreme x value is simply one that is particularly high or low. With multiple predictors, extreme x values may be particularly high or low for one or more predictors or may be "unusual" combinations of predictor values (e.g., with two predictors that are positively correlated, an unusual combination of predictor values might be a high value of one predictor paired with a low value of the other predictor).

Note that — for our purposes — we consider a data point to be an outlier *only* if it is extreme with respect to the other y values, not the x values.

A data point is **influential** if it unduly influences any part of regression analysis, such as the predicted responses, the estimated slope coefficients, or the hypothesis test results. Outliers and high-leverage data points have the potential to be influential, but we generally have to investigate further to determine whether or not they are actually influential.

- Leverage points by themselves are not bad, but along with the fact if the corresponding y is bad, then such leverage points are called influence points.

- way to determine influential point:

Find the best fitting line twice - once with the point included and once data point excluded. If the two lines vary too much then the point can be influential.

4.2 Influence, and Leverage Points

Recall that the hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

We usually focus on the diagonals of $\mathbf{H}$ so that

$$h_{ii} = x_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} x_i.$$

where $x_i^\top$ is the i th row of the $\mathbf{X}$ matrix. The h_{ii} is a standardized measure of distance of the i th x from the center of all the x s. Also recall that $\text{rank}(\mathbf{H}) = p + 1$. Thus, $n^{-1} \sum h_{ii} = (p + 1)/n$. So intuitively, any i whose h_{ii} is more than $2(p + 1)/n$ has abnormally large leverage.

Leverage points by themselves are not bad, but along with the fact if the corresponding y is bad, then such leverage points are called *influence* points. Thus, one must look at both the h_{ii} and the residuals.

Cook's Distance A simple question to ask is, how much is the regression estimator changing when I remove the i th observation. That is, we would like to measure the distance between $\hat{\beta}_{\text{OLS}}$ and $\hat{\beta}_{\text{OLS},(i)}$. This can be measured by the Cook's distance:

$$D_i = \frac{(\hat{\beta}_{\text{OLS},(i)} - \hat{\beta}_{\text{OLS}})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta}_{\text{OLS},(i)} - \hat{\beta}_{\text{OLS}})}{(p + 1) \text{MS}_{\text{Res}}} = \frac{r_i^2}{p + 1} \frac{h_{ii}}{1 - h_{ii}}.$$

If $D_i > 1$, then we say that the i th observation is an influential point.

No, not all influential points are outliers, and not all outliers are influential. An outlier is a point with a large residual (it is far from the trend of the data), while an influential point is one that has a large impact on the regression model itself, such as changing the slope or coefficients. A point can be an outlier without being influential if it has a high leverage, but the overall trend of the data is not significantly affected by its removal. Conversely, a point can be influential without being an outlier if it pulls the regression line in a new direction, even if it's not far from the rest of the data points (as shown in [this example from Medium](#)). 

	Outlier	Influential Point
Definition	A data point with a large residual, meaning it doesn't follow the trend of the other data points.	A data point that significantly impacts the regression model's coefficients, predictions, or hypothesis tests if removed.
Relationship to Data	Deviates significantly from the general pattern in the y-values.	Drastically alters the regression line or model's outcome.
Example	A point that is far from the regression line but does not significantly change the slope of the line.	A point that is not far from the data but has high leverage, causing it to pull the regression line towards itself.
Can be Both?	A point can be both an outlier and influential if it has a large residual and high leverage, causing a significant impact.	A point can be both influential and an outlier.

